

ZHALPAQTAL, KazakhstanWMO: **343980**Lat: **49.667N**Lon: **49.483E**Elev: **10**StdP: **101.20**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-22.8	-28.6	0.3	-25.6	-25.7	0.4	-22.6	11.7	-7.1	10.1	-8.5	3.1	50	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	37.1	20.0	35.4	19.3	33.7	18.8	21.3	32.9	20.6	31.8	19.9	30.5	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	23.7	17.1	12.2	23.4	16.1	11.4	24.0	61.7	33.1	59.1	31.9	56.8	30.4	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.3	7.3	-27.4	39.5	3.6	1.7	-30.0	40.7	-32.1	41.7	-34.2	42.7	-36.8	43.9		
(5)			-27.5	23.1	3.6	1.8	-30.1	24.4	-32.2	25.4	-34.2	26.4	-36.8	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-9.7	-9.4	-1.1	9.9	18.1	22.6	25.1	24.1	16.7	8.1	0.6	-6.1	(6)
(7)		DBStd	13.62	6.93	6.82	5.98	4.78	4.67	4.39	3.44	4.20	4.34	4.68	5.40	6.46	(7)
(8)		HDD10.0	2442	611	543	346	61	2	0	0	0	3	93	284	500	(8)
(9)		HDD18.3	4277	869	777	603	257	64	13	1	7	80	316	533	758	(9)
(10)		CDD10.0	1831	0	0	1	56	252	377	469	437	204	35	1	0	(10)
(11)		CDD18.3	624	0	0	0	2	55	139	212	184	31	1	0	0	(11)
(12)		CDH23.3	7799	0	0	0	25	692	1745	2598	2333	398	7	0	0	(12)
(13)		CDH26.7	3840	0	0	0	3	282	854	1327	1234	139	1	0	0	(13)
(14)	Wind	WSAvg	3.7	4.1	4.1	4.0	4.1	3.8	3.7	3.4	3.3	3.4	3.5	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	243	23	18	20	20	19	26	22	8	27	25	21		(15)
(16)		PrecMax	352	61	52	70	66	87	107	107	85	64	61	81	48	(16)
(17)		PrecMin	110	4	0	0	0	0	0	1	0	0	4	2	1	(17)
(18)		PrecStd	62	12	13	18	17	18	24	23	18	15	16	19	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	3.7	16.2	26.4	34.5	38.1	39.5	38.8	34.0	24.1	14.1	7.1	(19)
(20)			MCWB	3.7	2.4	9.6	16.7	17.6	19.5	20.9	20.6	17.8	12.9	10.6	6.0	(20)
(21)		2%	DB	2.2	2.2	11.9	23.2	32.2	35.7	37.2	37.0	30.3	21.2	10.8	4.6	(21)
(22)			MCWB	1.6	1.1	7.1	13.8	17.3	19.0	20.1	20.2	16.7	12.2	8.6	3.5	(22)
(23)		5%	DB	0.8	1.1	8.9	20.4	30.1	33.9	35.2	35.2	27.9	18.3	9.1	2.7	(23)
(24)			MCWB	0.2	0.3	6.0	12.7	17.2	18.9	19.7	19.1	15.7	11.2	7.4	1.8	(24)
(25)		10%	DB	-0.1	0.2	6.2	18.1	27.4	31.7	33.2	33.2	25.3	15.7	7.4	1.1	(25)
(26)			MCWB	-0.5	-0.4	3.7	11.2	16.3	18.1	19.0	18.7	14.8	10.2	6.1	0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	2.7	10.1	17.6	19.8	22.0	22.9	22.2	19.5	15.6	11.9	6.3	(27)
(28)			MCDWB	4.5	3.5	15.5	25.4	29.8	32.9	32.2	34.0	30.3	19.5	13.7	6.8	(28)
(29)		2%	WB	1.7	1.3	7.8	15.2	18.5	20.8	21.4	21.0	17.9	13.5	9.2	3.9	(29)
(30)			MCDWB	2.0	1.9	11.1	21.5	29.0	31.7	32.7	33.9	26.6	18.6	10.5	4.8	(30)
(31)		5%	WB	0.4	0.4	6.0	13.5	17.6	19.8	20.7	20.2	16.7	12.2	7.6	2.0	(31)
(32)			MCDWB	0.8	1.0	9.1	18.9	27.7	29.9	31.5	32.3	25.2	17.1	8.8	2.6	(32)
(33)		10%	WB	-0.6	-0.4	3.9	12.0	16.7	18.9	20.0	19.5	15.7	10.8	6.1	0.5	(33)
(34)			MCDWB	-0.1	0.3	5.7	17.2	26.1	28.9	30.2	30.9	23.8	14.8	7.3	1.1	(34)
(35)	Mean Daily Temperature Range		MDBR	6.6	7.4	7.8	10.6	12.2	12.6	12.3	13.2	12.5	10.3	6.3	5.5	(35)
(36)		5% DB	MCDBR	4.5	5.6	11.0	14.0	15.1	15.3	14.5	15.7	16.4	14.9	8.4	4.3	(36)
(37)			MCWBR	4.5	5.1	7.1	6.8	5.4	5.1	4.8	5.6	7.0	8.1	6.5	3.9	(37)
(38)		5% WB	MCDWB	4.7	5.4	10.8	12.8	13.6	13.3	12.6	13.8	13.8	12.4	7.3	3.9	(38)
(39)			MCWBR	4.7	5.1	7.1	6.5	5.7	4.9	4.5	5.4	6.4	7.7	6.1	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.350	0.394	0.452	0.430	0.410	0.417	0.430	0.382	0.363	0.341	0.326	(40)	
(41)		taud	2.069	2.027	2.085	2.034	2.177	2.300	2.286	2.223	2.322	2.302	2.254	2.135	(41)	
(42)		Ebn at Noon	667	756	796	789	830	850	838	804	809	750	666	614	(42)	
(43)		Edh at Noon	87	117	132	155	140	125	125	127	104	89	73	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.10	2.24	3.40	4.65	6.11	6.60	6.40	5.57	3.98	2.31	1.09	0.74	(44)	
(45)		RadStd	0.25	0.42	0.48	0.34	0.44	0.55	0.43	0.33	0.35	0.29	0.16	0.14	(45)	